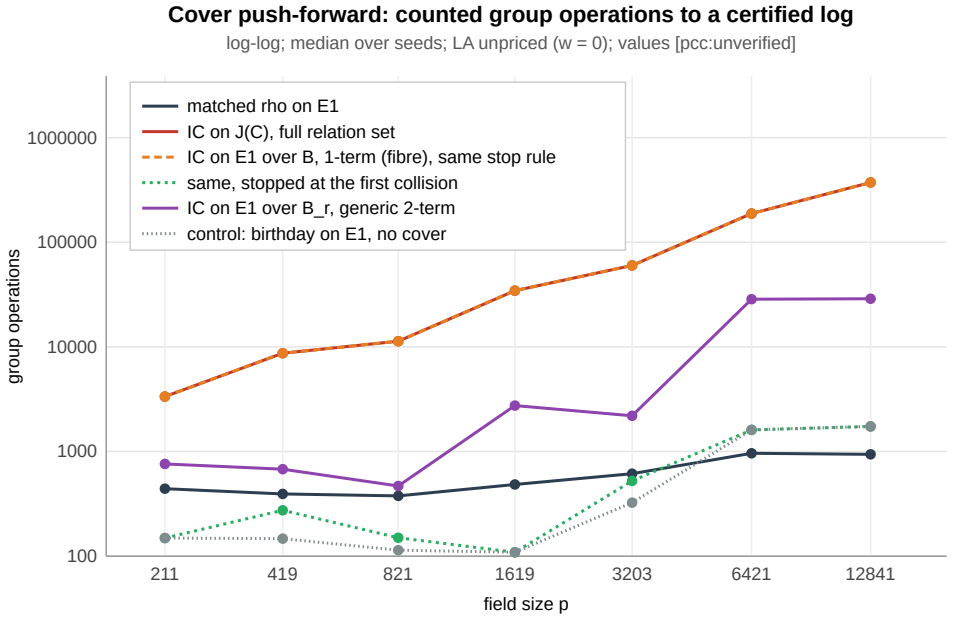
Cover mechanism and endomorphism census

Cover mechanism and endomorphism census (2026-11-12)

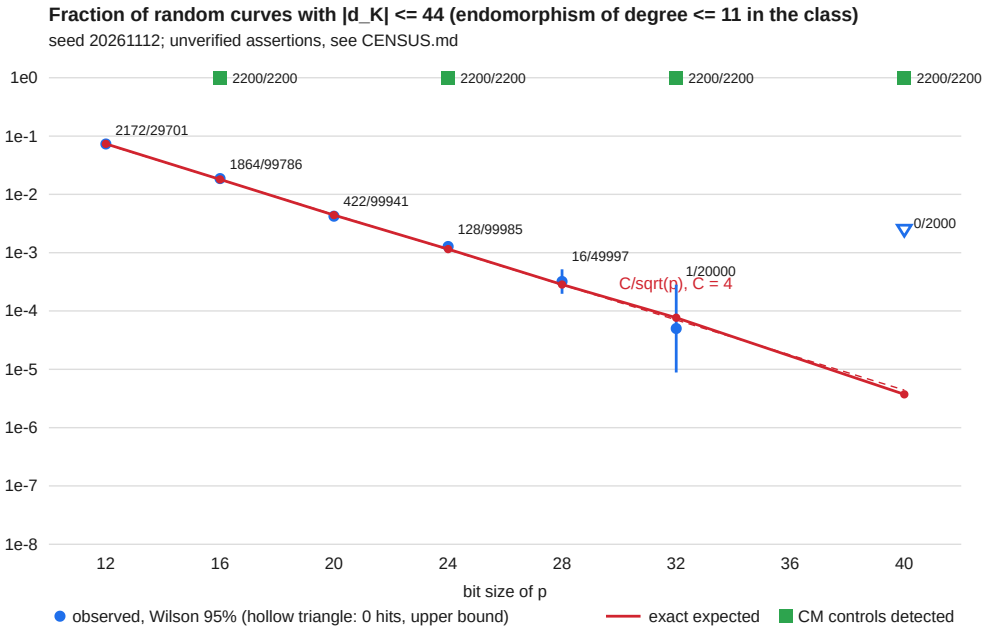
All values `[pcc:unverified]`.



Target-side factor base B = phi1(degree-1 places of C)

p	|B|/#E1	|B cap subgroup|/r	places per image (2 : 1)	pushed rows valid E1 / E2
211	0.5367	0.5505	115 : 2	372/372 / 372/372
419	0.4934	0.4071	221 : 2	693/693 / 693/693
821	0.4707	0.4789	399 : 2	1227/1227 / 1227/1227
1619	0.4819	0.4638	771 : 2	2340/2340 / 2340/2340
3203	0.4904	0.5112	1533 : 2	4626/4626 / 4626/4626
6421	0.4922	0.4951	3139 : 2	9447/9447 , 9447/9447
12841	0.5032	0.5080	6377 : 2	19161/19161 , 19161/19161

Cover push-forward



Endomorphism-degree census

Cover push-forward: what a bielliptic cover gives on the embedded elliptic subgroup

Protocol (frozen before data, with two dated amendments): PROTOCOL.md, hashes in `frozen.sha256`. Code: `src/cryptanalysis/chained_transfer/pushforward.rs`, `src/bin/cover_pushforward.rs`. Data: `pushforward.json` (primary run, 3 seeds per prime), `pushforward_15seeds.json` (supplementary, 15 seeds), `run.log`, `pushforward.svg`. Bench record: `benchmarks/records/20261112T1550Z-cover-pushforward.json` (left open). Every number below is a counted group operation or a count from those files and carries [pcc:unverified]. Counts are deterministic and exact; no timing claim is made.

Verdict against the frozen rule

The cover supplies no logarithm on the embedded elliptic subgroup and no factor base smaller than the group. [pcc:unverified] All three conditions of the frozen rule hold at all 7 primes (211 to 12841): (1) 37,866 relations were pushed through `phi1_*` and `phi2_*`, 0 were invalid on `E1` and 0 on `E2`, and 0 pushed `E2` relations involve the target; (2) `|B cap <G_E>| / r` lies between 0.4071 and 0.5505, never below 0.4; (3) IC on `J(C)`, IC on `E1` over `B` with the same stop rule, and the generic two-term IC on `E1` over `B_r` each cost more than the matched rho on `E1` at every prime and at all three prices of the unpriced linear algebra (gap ratios 1.24 to 43,658 depending on arm and price, tables below). [pcc:unverified]

The owner's sentence that "IC on `J(C)` should be exactly two-term IC on `E1` over `B`" is half right and half wrong, and the wrong half is the sharper result: the pushed relation is not an independent two-term decomposition. It is the degenerate `[2]Q = Q + Q`, so the genus-2 index calculus on the pulled-back instance is a collision search on `E1` that discards half its trials. [pcc:unverified]

The mechanism, stated precisely

Notation: `C : y^2 = F(x^2)`, `phi1(x, y) = (x^2, y)`, `iota(x, y) = (-x, y)`, `G = phi1^*(G_E)`, `T = phi1^*(T_E)`, `B = phi1(affine degree-1 places of C)`.

1. **`B` is the half of `E1` with square abscissa.** A point `(X, Y)` of `E1` has a rational preimage exactly when `X` is a square in `F_p`. So `|B|` is the number of affine points of `E1` with `X` a square or zero, about `#E1 / 2`. Each image has two places over it, `x` and `-x`, except `X = 0`, where `x = 0` is one place (ramified). Measured: `|B| / #E1` between 0.4707 and 0.5367 over the 7 primes, the count of `B` from the places of `C` equal to the count from `E1` alone at every prime, fibre sizes only 2 and 1, with exactly 2 images of fibre size 1 (the two points with `X = 0`) at every prime. [pcc:unverified]
2. **The reduced class of `[alpha]G + [beta]T` is a fibre.** `Bielliptic::pull(Q)` returns the divisor `P + iota(P)` of the two preimages of `Q`. The reduced Mumford representative of a degree-2 class is unique, and `phi1^*` is a homomorphism, so `[alpha]G + [beta]T = phi1^*(Q)` with `Q = [alpha]G_E + [beta]T_E` is that fibre. It splits into degree-1 places exactly when `X(Q)` is a nonzero square (probability about 1/2), and then its two places `P, iota(P)` have the same image `phi1(P) = phi1(iota(P)) = Q`. Measured: 37,866 of 37,866 relations collected by the genus-2 IC are of this fibre type (primary run), and 189,330 of 189,330 in the 15-seed run. [pcc:unverified]
3. **Pushed through `phi1`, the relation is `[2]Q = Q + Q`.** The pushed relation `phi1(P1) + phi1(P2) = [2 alpha]G_E + [2 beta]T_E` holds in every case (0 invalid in 37,866 and in 189,330), but its two terms are the same point of `B`. It carries one factor-base element, not two.
4. **The matrix is a collision table.** Every J(C) row has two entries of equal sign, on the columns of `x` and `-x`. The columns actually used are exactly twice the columns of the native `E1` arm (58 against 29 at p = 211 and 1,608 against 804 at p = 12841, medians; the factor 2 holds at every seed of both runs), and a row can only combine with another row on the same `+-Q`. The elimination returns the log through a repeated `X(Q)`, that is a birthday collision on `<G_E>`. The cover contributes the acceptance test "`X(Q)` is a square", which discards about half of the trials. [pcc:unverified]
5. **Through `phi2` the pushed relation is trivial.** `phi2_* phi1^* = 0`: for every relation `phi2_*(G)` and `phi2_*(T)` are the identity, so the pushed `E2` relation reads `sum = 0` whatever `alpha, beta` are (0 of 37,866 rows have a non-identity `phi2_*(G)` or `phi2_*(T)`; the two places of a row never have equal `phi2` images, 0 of 37,866). It holds (0 invalid) and says nothing about the target. [pcc:unverified]

No step computes a logarithm on the embedded subgroup other than by finding two equal multiples `Q = +-Q'` inside `<G_E>` itself, which needs no cover. The control arm (A6) makes this explicit: the same stream on `E1` with no filter finds the same collisions at or before the cover arms (below).

Results

Primary run, 3 seeds per prime, seed 0 reproduces the `results_v3.json` rows (IC on J(C) group ops 3,419 and 8,475 at p = 211, 419; rho 428 and 357); the copy of the relation collector reproduces the library's `genus2_index_calculus` exactly (ops, trials, relations, recovered log, LA multiplications) at the 6 primes up to 6421 where it was cross-checked. All 7 x 3 x 7 arm results were certified by an independent double-and-add. [pcc:unverified]

Factor base and pushed relations (all cells [pcc:unverified])

p	r	#E1	abs B	B / #E1	B∩ / r	places per image 2 : 1	X=0 images	pushed rows valid E1	valid E2	E2 rov carryi targ
211	109	218	117	0.5367	0.5505	115 : 2	2	372 / 372	372 / 372	
419	113	452	223	0.4934	0.4071	221 : 2	2	693 / 693	693 / 693	
821	71	852	401	0.4707	0.4789	399 : 2	2	1227 / 1227	1227 / 1227	

p	r	#E1	abs B	B / #E1	B∩ / r	places per image 2 : 1	X=0 images	pushed rows valid E1	valid E2	E2 rov carryi targ
1619	401	1604	773	0.4819	0.4638	771 : 2	2	2340 / 2340	2340 / 2340	
3203	313	3130	1535	0.4904	0.5112	1533 : 2	2	4626 / 4626	4626 / 4626	
6421	3191	6382	3141	0.4922	0.4951	3139 : 2	2	9447 / 9447	9447 / 9447	
12841	3169	12676	6379	0.5032	0.5080	6377 : 2	2	19161 / 19161	19161 / 19161	

Comparison per prime (medians over 3 seeds, group operations; all cells [pcc:unverified])

IC J(C) and IC E1 fibre use the same random stream and the same stop rule (relations = J(C)'s). The two-term arm builds `B_r` by `r - 2` counted additions (in parentheses) and is charged one counted subtraction per candidate.

p	r	rho E1	IC J(C) ops	trials	rel	trials/rel	IC E1 fibre (same stream) ops	IC E1 two-term ops (base build)	trials/rel	rel
211	109	441	3354	220	124	1.77	3354	761 (107)	1.00	38
419	113	393	8694	567	231	2.45	8694	679 (111)	1.00	31
821	71	377	11353	866	409	2.12	11353	469 (69)	1.00	25
1619	401	485	34521	1653	780	2.12	34521	2754 (399)	1.01	101
3203	313	614	60059	3062	1542	1.99	60059	2194 (311)	1.00	88
6421	3191	963	188130	6331	3149	2.01	188130	28546 (3189)	1.00	798
12841	3169	942	372157	12528	6387	1.96	372157	28897 (3167)	1.00	813

Gap to matched rho at the three prices of linear algebra (median total / rho, w = 0 / 0.1 / 1 GAE per `Z/r` multiplication; all cells [pcc:unverified])

p	IC J(C)	IC E1 fibre	IC E1 two-term
211	7.61/10.92/40.79	7.61/8.48/16.32	1.73/2.63/10.37
419	23.74/37.72/156.75	23.74/25.36/39.19	1.73/2.43/8.79
821	30.50/73.76/467.20	30.50/32.54/50.90	1.24/1.68/5.55
1619	70.50/194.98/1315.29	70.50/85.78/223.29	5.68/16.49/113.80
3203	97.13/482.88/3954.64	97.13/117.47/300.55	3.62/8.48/56.07
6421	198.21/1225.97/10475.79	198.21/451.96/2735.72	29.47/446.60/4200.83

p	IC J(C)	IC E1 fibre	IC E1 two-term
12841	393.59/4720.07/43658.40	393.59/940.08/5858.48	30.37/485.52/4585.04

The gap is the distance remaining. In the primary run no arm is below rho at any price in any prime. In the 15-seed run the two-term arm was below rho on the same seed in 1 of 15 seeds at p = 821 (472 against 475 on that seed), and the full-stop arms in 0 of 105. [pcc:unverified]

Stopping at the first collision (supplementary run, 15 seeds; group operations; all cells [pcc:unverified])

This is the J(C) relation list (equivalently the native `E1` fibre arm; the two have identical group-operation counts at every seed) read at the first repeated `X(Q)`, against the control that uses no cover.

p	r	rho E1 median (range)	J(C) IC stopped at first collision median (range), seeds below rho	E1 birthday control median (range), seeds below rho
211	109	425 (310-516)	221 (47-548), 13/15	149 (47-280), 15/15
419	113	417 (327-519)	231 (45-491), 14/15	147 (45-261), 15/15
821	71	308 (240-475)	114 (40-250), 15/15	102 (40-218), 15/15
1619	401	559 (382-949)	543 (81-1512), 7/15	448 (81-665), 13/15
3203	313	517 (384-888)	524 (111-728), 7/15	325 (111-644), 11/15
6421	3191	925 (690-1970)	2254 (652-4694), 2/15	1644 (413-3090), 3/15
12841	3169	876 (584-2649)	1699 (751-5953), 3/15	1176 (587-2929), 6/15

The cover arm is at or above the control in every seed by construction (it accepts a subset of the control's trials). At the three smallest primes both are below the median rho (the cover arm below rho on 13, 14 and 15 of 15 seeds, the control on 15 of 15 at each). At p = 6421 (r = 3191) the cover arm is above rho on 13 of 15 seeds and the control on 12 of 15; at p = 12841 (r = 3169) 12 of 15 and 9 of 15. [pcc:unverified] Rho's counted cost contains a fixed start-up (sixteen jump points, each built from two scalar multiplications; not measured separately here), which is the likely reason a birthday search wins at small `r`; the trend in `r` (birthday search about `sqrt(r)` trials of two scalar multiplications each, rho about `sqrt(r)` steps of one addition) is the expected one, not a property of the cover. As a measured gap to the IC baseline on `J(C)`, stopping at the first collision costs between 0.46% (p = 12841) and 6.33% (p = 211) of the operations of the full relation set (median over 15 seeds). It uses no genus-2 structure beyond the acceptance filter, and the control without the filter is never worse. [pcc:unverified]

Local wall-clock ratio of one Cantor addition to one elliptic addition in this implementation (unisolated host, no timing claim): 15.8 to 24.5 over the 7 primes. [pcc:unverified] The J(C) and E1-fibre arms have identical group-operation counts, so on an equal-operation basis `E1` is the cheaper place to run the same search. The transfer (cover, `phi^*`, instance) is a one-time cost of 0.3 to 8.9 ms per curve and is in no total.

Prediction against outcome

id	frozen prediction (probability)	outcome
P1	`|B|/#E1` in [0.45, 0.55] (0.90); `|B cap <G_E>|/r` in [0.40, 0.60] (0.85)	held: 0.4707 to 0.5367 and 0.4071 to 0.5505
P2	two places per image except `X = 0` (0.97)	held at all 7 primes
P3	pushed relations valid on `E1` and on `E2` (0.98, 0.97)	held: 0 invalid in 37,866 and in 189,330
P4	`phi2_*` of generator and target are the identity, 0 pushed `E2` rows carry the target (0.98)	held: 0 rows
P5	every genus-2 IC relation is fibre type (0.95)	held: 100%
P6	J(C) IC and two-term E1 IC have the same trials per relation and relation count (0.15)	did not hold, as expected: trials per relation 1.77 to 2.45 against 1.00 to 1.01, relations 3.3 to 17.5 times as many on J(C)
P7	A1 and A4 accept identical trials (0.93)	held at every seed of both runs, and identical group operations
P8	full-stop arms above rho everywhere and at all prices (0.92); A4 first collision below rho at some cell (0.25)	first held on the primary run; second occurred (primary-run median below rho at p = 211 to 3203)
P9	the two-term base build alone exceeds rho in at least one cell (0.60)	held at p = 6421 (3,189 against 963) and 12841 (3,167 against 942)
A6	control first collision below the cover arm's at every cell (0.85)	at or below in all 7 cells, strictly below in 3 of 7 medians on the primary run (equal at 211, 1619, 6421, 12841 where the filter did not delay the first collision), strictly below in 7 of 7 on the 15-seed run; judged held on the weak reading and not on the strict wording of the frozen text

All values [pcc:unverified].

What a cover would have to do differently

The owner's rule is that a shortcut must say how it computes logarithms on the embedded elliptic subgroup. For the degree-2 bielliptic covers measured here the answer is: it does not, because the relations it produces on `J(C)` are images of single points of `<G_E>`, and the only information they contain is which multiples of `G_E` have a square abscissa. To help, a cover has to supply all of:

1. an explicit, evaluable map from the target's group into a place where relations can be found (here `phi^*`, supplied, and one-time);

2. a target-side factor base of size well below `r` (the bar: about `r^(1/2)` times a polylog, so that relation collection with a constant number of group operations per relation stays below the `sqrt(r)` of rho), on which relations are found at a constant rate;
3. relations whose terms are independent elements of that base. Here the base is half the group (`|B cap <G_E>| / r` about 0.5) and the two terms coincide.

A smaller Prym dimension does not enter: `E1` carries the whole discrete-log instance (`phi1_* phi1^* = [2]`), the Prym (the `E2` side here) is killed by `phi1_*`, and nothing computed on it can see the target. [pcc:unverified]

Limits and what went wrong

- One bielliptic curve per prime (the first candidate of the bench's search loop), weight-2 relations only, degree-2 covers only, `p` up to 12841. Nothing here speaks for covers of higher degree or non-split Pryms.
- The independence of the push check: the `E1` side compares the place sum with `[2 alpha]G_E + [2 beta]T_E` computed on `E1` alone; the `E2` side compares it with `phi2_*` as computed by `Bielliptic::push`, whose formula `place_image` shares with `phi` (both written from the same model). The vanishing `phi2_* phi1^*` is also a unit test of `bielliptic.rs`.
- A smoke run (2 primes, no record) preceded the full run and led to Amendment 1 (control arm A6); the primary run produced noisy first-collision counts and led to Amendment 2 (15 seeds). Both are post-hoc to the first data and labelled as such in the protocol. The verdict rule was judged on the primary run as frozen.
- The strict wording of the A6 prediction did not hold on the primary run (above).
- `cargo clippy --release --all-targets -- -D warnings` fails for the crate for reasons outside these files: two `manual_is_multiple_of` errors in `endo_census.rs` and one `unnecessary_map_or` error in `koblitz_index_calculus.rs`; this crate's `pushforward.rs` and `cover_pushforward.rs` are clean.

Endomorphism-degree census (2026-11-12)

Protocol: `PROTOCOL.md`, frozen by hash before any data (`frozen.sha256`, re-verified after the run), with one dated reporting amendment in `AMENDMENTS.md`. Code: `src/bin/endo_degree_census.rs`, `src/cryptanalysis/chained_transfer/endo_census.rs`. Bench record: `benchmarks/records/20261112T1558Z-endo-degree-census.json` (open). Every number below is an unverified assertion produced by that run (`[pcc:unverified]`); none is signed by pcc. Nothing here is a claim about index-calculus cost.

Verdict against the frozen rule

- Rule 1 (validity) holds: all 8800 control curves are detected with the right `d_K` and the right crater status, the 8 unit tests pass, and the largest `|z|` over the tested null sizes is 1.447 [pcc:unverified].
- Rule 2 fires **as written**: 185 registry rows of (a) have `p >= 2^30`, `|d_K| <= 44`, `d_K` not in {-3, -4}, and are not known non-crater (141 distinct `(p, t)`) [pcc:unverified]. The program prints `WORTH_RUNNING_ON_NATURAL_CURVES`.
- Read what those rows are. 184 of the 185 are decidable and all 184 are on the crater [pcc:unverified]; they are the lab's own CM-constructed curves (for example `D = -19` with a Frobenius order of large conductor), not curves anyone deployed. Restricted to (b), the named and published curves, the only candidate is `bls/Bandersnatch` (`d_K = -8`, class minimum degree 2, 255 bits; its crater status is "unknown" because the registry row records no `a`, `b`) [pcc:unverified]. The honest summary is outcome 3 in spirit: beyond `j = 0` and `j = 1728`, curves with a degree `<= 11` endomorphism exist **only by construction**, and the registry already holds such curves for `D = -7, -8, -11, -19, -43` at `p` up to 61, 255, 60, 170, 61 bits (counts below) [pcc:unverified]. The small-`D_E` experiment does not need any new curve construction at the sizes a solver can reach.

Null population: the prediction was right

Uniformly random `(a, b)` over random primes of exactly `b` bits, seed 20261112, order by the crate's Shanks-Mestre counter; zero counter failures at every size [pcc:unverified]. "Expected" is the exact formula (sum over the realised primes), frozen in `predict.py` before the run.

bits	curves (ordinary)	hits `|d_K| <=44`	fraction	expected hits	frozen prediction	z
12	30000 (29701)	2172	7.313e-2	2206.7	7.15e-2	-0.78
16	100000 (99786)	1864	1.868e-2	1803.3	1.88e-2	1.45
20	100000 (99941)	422	4.222e-3	442.3	4.29e-3	-0.97
24	100000 (99985)	128	1.280e-3	115.1	1.157e-3	1.20
28	50000 (49997)	16	3.200e-4	14.2	3.14e-4	0.47
32	20000 (20000)	1	5.0e-5	1.54	6.98e-5	-0.43
40	2000 (2000)	0	0 (upper 95%: 1.9e-3)	0.007	4.03e-6	-0.09

[pcc:unverified] (every row). Sizes 12 to 28 are "tested" (expected hits at least 5) and agree with the exact formula; 32 and 40 bits are consistent but have no power, as the protocol said. The exact formula was also checked against an exhaustive enumeration of every `(a, b)` for `p = 101, 103` (unit test) and `p = 101, 211` (`predict.py` development) [pcc:unverified]. Scaling: `P(|d_K|<=44) ~ C/sqrt(p)`, `C` near 4 for `p` around 1.5 times `2^(b-1)` (ratios 3.7 to 4.5 across sizes) [pcc:unverified]. For a 256-bit curve that is about `4 * 2^-128`. See `census.svg`.

Per-discriminant counts of the 2172 hits at 12 bits and 1864 at 16 bits are in `census.json` (`hit44_by_d`); `d_K = -3` and `-4` carry the most, as the class numbers and `1/sqrt|d|` weights predict [pcc:unverified].

Degree buckets, null (minimum endomorphism degree over the isogeny class)

bits	1	2	3	4-5	6-8	9-11	12-1100	> 11, unresolved
12	659	369	176	338	273	357	15905	11624
16	549	340	136	297	243	299	15320	82602
20	129	65	24	70	47	87	3774	95745
24	45	24	7	22	17	13	1006	98851
28	6	3	1	3	2	1	134	49847
32	0	0	0	0	0	1	16	19983
40	0	0	0	0	0	0	0	2000

[pcc:unverified]. Counts are over ordinary curves; the buckets "12-1100" and "> 11" come from a search to `|d_K| <= 4400` (so "> 11" at these sizes means `|d_K| > 4400`, minimum degree above 1100). Degree 1 is an automorphism (`d_K = -3, -4`).

The class bound is not the curve: the crater (post hoc)

The census reports the **class** minimum degree. For a given `E`, `End(E)` has conductor `c | f`, and its minimum degree is about `c^2 |d_K| / 4` unless `c = 1`. The crater is decided exactly only for `h(d_K) = 1`. Among null hits, the crater is rare [pcc:unverified]:

bits	hits found on the crater (decidable only for `h(d_K)=1`)	`f / sqrt(p)` over hits (median)
12	26 of 2172 hits	0.445
16	7 of 1864	0.450
20	0 of 422	0.444
24	0 of 128	0.480
28	0 of 16	0.516

[pcc:unverified]. `posthoc.py` (written after the data, labelled post hoc) computes the exact probability that `End(E) = O_K`: with the same Hurwitz derivation it is the `c = 1` term only. For `h = 1` discriminants it gives 1.196e-3, 7.74e-5, 4.62e-6, 2.7e-7 at 12, 16, 20, 24 bits, i.e. 35.5, 7.7, 0.46 and 0.03 expected crater hits against 26, 7, 0, 0 observed [pcc:unverified]. It falls by a factor of about 16 per 4 bits: **`P(E on the crater) ~ 1/p`**, not `1/sqrt(p)`. The exact probability that `|D_E| <= 44` (orders of conductor `c | f` with `|d_K| c^2 <= 44`) is about three times the crater probability (3.71e-3 at 12 bits, 2.38e-4 at 16 bits) [pcc:unverified].

So for a random curve, the class has a degree `<= 11` endomorphism with probability near `4/sqrt(p)`, but `E` itself has one with probability near `1/p`. In a class-level hit, the isogeny that reaches the crater has degree `c`, and `f` is about `0.45 sqrt(p)` at the median, so the walk is as long as `sqrt(p)`-scale, the size of a rho run. This is the owner's point in numbers: a small-degree isogeny supplies no endomorphism unless the cycle closes, and here the cycle closes only after a `sqrt(p)`-scale descent.

Registry (a) and named curves (b)

Skipped, with reason: 102 binary, 68 Koblitz, 1371 extension-field and 4 subfield rows (1545 of 22278 rows, none prime-field); no prime row failed the Hasse check, so 20733 prime rows were analysed, 3 supersingular [pcc:unverified]. The registry is **not random**: 938 distinct `(p, t)` carry the 20733 rows [pcc:unverified], and it is built from volcano, conductor-gap, BN/BLS and testbed curves chosen for earlier experiments.

population	rows (ordinary)	`|d_K|<=44`	fraction
a: all prime rows	20733 (20730)	2184	10.5%
a: distinct `(p, t)`	938 (935)	273	29.2%
a: `p <= 16` bits	198 (195)	20	10.3%
a: 17-32 bits	19073	1012	5.3%
a: 33-64 bits	1161	1102	94.9%
a: 65-128 bits	15	11	73.3%
a: 129-256 bits	215	18	8.4%
a: > 256 bits	71	21	29.6%
a: `p >= 2^30`	1952	1152	59.0%
b: named standard	54	6	11.1%
b: pairing-friendly (BN, BLS)	29	29	100%
b: other named (SafeCurves, Pallas, …)	40	4	10.0%

[pcc:unverified] (every row). The 33-64 bit tier is 95% hits because it holds the lab's CM-constructed families; that is a description of the registry, not an estimate for any random curve. For comparison, the exact null expectation at 24 bits is 1.15e-3 [pcc:unverified].

Hits by discriminant for `p >= 2^30` (rows / distinct `(p,t)`): -3: 173 / 52; -4: 79 / 3; -7: 232 / 7; -8: 139 / 3; -11: 74 / 4; -19: 403 / 118; -43: 52 / 15 [pcc:unverified]. Bit ranges of the non-(-3,-4) families: -7: 46 to 61 bits, -8: 53 to 255, -11: 47 to 60, -19: 37 to 170, -43: 42 to 61 [pcc:unverified]. Note few distinct `(p, t)` for -4, -7, -8 and -11 (3, 7, 3, 4): many rows are other models of the same few classes.

Named curves (orders taken from `docs/curves/registry.json`, whose standards rows cite `docs/curves/standards/registry.json`, `src/ecc/curve.rs` and `src/ecc/curve_zoo.rs`; the P-256 and secp256k1 `p` and `n` literals were also found verbatim in `src/ecc/curve.rs` and agree with the registry row) are [pcc:unverified]:

- `d_K = -3`, on the crater (`j = 0`, automorphism of order 6): secp160k1, secp192k1, secp224k1, secp256k1, wap-wsg-idm-ecid-wtls8, -wtls9; every BN curve (bn158 to bn638, Fp224BN to Fp512BN, bn254, Fp254BN) and every BLS12, BLS24 and BLS48 G1 curve in the registry; Pallas, Vesta, Tweedledum and Tweedledee. That is 6 + 28 + 4 = 38 rows.
- `d_K = -8`, class minimum degree 2: `bls/Bandersnatch` only.
- No named standard curve has `d_K = -4`, and none has `-7, -11, -15, ..., -43`. The rest of the 54 named standard curves (P-192 to P-521, brainpool, Curve25519 and Curve448 and their forms, GOST, SM2, FRP256v1, prime239vN, secp112 to 128 and 160r) have `|d_K| > 44`, so their class has no endomorphism of degree `<= 11`; the class minimum degree is not computed (needs the factorisation of `Delta`) [pcc:unverified].

Controls

Positive: `D = -3, -4, -7, -8, -11, -19, -43`, 200 curves each at 16, 24, 32 and 40 bits with a random twist: 5600 of 5600 detected with `d_K = D`, all on the crater. Negative (non-maximal): `D = -12, -16, -27, -28`, same sizes: 3200 of 3200 detected with `d_K = -3, -4, -3, -7` and all reported **not** on the crater; counter failures 0 [pcc:unverified]. (Per-`D`, per-size rows are in `census.json` and `run.log`.)

What this implies for `small_de_endo_fb_20261112`

1. **How often does a low-degree endomorphism exist?** On a random curve, in the isogeny class, with probability near `4/sqrt(p)` (`d_K` within 44); on the curve itself, near `1/p` [pcc:unverified]. For cryptographic `p` neither is observable. Every named curve with a degree `<= 11` endomorphism in the repository other than `d_K = -3` is `bls/Bandersnatch`, designed for it.
2. **Beyond `j = 0` and 1728, the experiment is a designed-curve experiment.** It must run on CM-constructed curves (one-time construction cost, never in a per-instance total), and the registry already has them for `D = -7, -8, -11, -19, -43` at 37 to 61 bits and `D = -8` at 255 bits [pcc:unverified]. A claim from it applies to a designer choosing such a curve, not to deployed standard curves.
3. **Choose `p` with `f` small.** A class-level hit on a random curve needs a `sqrt(p)`-scale descent; use curves with `End(E) = O_K` (decided exactly by `j(E) = j_D mod p` for `h = 1`, as in this program), so there is no isogeny cost.
4. The census says nothing about whether a factor base closed under the endomorphism is of useful size; the experiment's own decision rule is untouched.

Limits

- The crater is decided only for `h(d_K) = 1` (seven discriminants); for the other eight the repository has no Hilbert class polynomial, so those rows are "class bound only".

- Ten registry rows have no recorded `a` or `b` (crater cannot be computed); Bandersnatch is one of them.
- The null's counter could give up (twist retry, then count as failed): 0 failures. The counter is the crate's `frobenius_trace_bsgs`; sampling is deterministic (`null_transcript_sha256` in `census.json`), and two runs gave byte-identical `rows.jsonl` and `census.json` apart from the elapsed time [pcc:unverified].
- 32 and 40 bits have almost no expected hits; they do not test the prediction.
- Ran locally on 4 cores in 104 seconds, not on the EKS Spot route; this is a small CPU job, and the task asked for the local `./target/release` binary. The bench record is left open.

Files

`PROTOCOL.md`, `frozen.sha256`, `AMENDMENTS.md`, `predict.py`/`predict.json` (pre-data), `census.json`, `rows.jsonl` (hit and named rows), `run.log`, `census.svg` (`plot.py`), `posthoc.py`/`posthoc.json` (post hoc). Reproduce: `cargo build --release --bin endo_degree_census && ./target/release/endo_degree_census`.